

JOSÉ LUIS CARMONA JIMÉNEZ

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Nationality: Spanish | Age: 31 | Address:

WORK EXPERIENCE

From: 16/09/2024 To: Current	Postdoctoral Researcher in the Simion Stoilow Institute of mathematics of the Romanian academy (IMAR). This Postdoctoral position is supported by the PNRR-III-C9-2023-I8 grant CF 149/31.07.2023 Conformal Aspects of Geometry and Dynamics in Bucharest.	IMAR
From: 01/09/2022 To: 10/09/2024	Teacher in a secondary and upper-secondary School. Permanent position.	
From: 01/10/2020 To: 31/08/2022	Predoctoral Researcher in differential geometry University Complutense of Madrid (UCM). FPI predoctoral fellow at the group of research Geometrical, topological and combinatorial structures associated with dynamical systems. Principal researcher: Marco Castrillón López.	UCM
From: 01/07/2020 To: 30/09/2020	Researcher in differential geometry Instituto de Ciencias Matemáticas (ICMAT). Researcher grant at ICMAT (CSIC) associated with the research group ALGEBRA AND GEOMETRY.	ICMAT

EDUCATION

10/05/2024	Ph.D. in Mathematics Ph.D. Thesis entitled "Homogeneous descriptions and families of homogeneous structures". Department of Algebra, Geometry and Topology in University Complutense of Madrid. Advisor: Marco Castrillón López.	University Complutense of Madrid
2019	Master's in Secondary Mathematics Education at University of Granada.	University of Granada
2018	Master of Science in advanced mathematics. GPA: 9.58/10. Dissertation: "The Ambrose-Singer Theorem". Tutor: Marco Castrillón López.	University Complutense of Madrid
2017	Degree in Mathematics GPA: 9.17/10.	University of Málaga

RESEARCH INTERESTS:

Differential Geometry, Geometric Analysis, Homogeneous Geometry, Complex Hyperbolic Geometry, Invariant Connections, Submanifold Geometry.

PUBLICATIONS

ACCEPTED:

- [Car26] Carmona Jiménez, J.L. *On Weyl structures reducible in the direction of the Lee form*. Proc. Amer. Math. Soc. (2026). In press, arXiv: [10.48550/arXiv.2507.17496](https://arxiv.org/abs/10.48550/arXiv.2507.17496)
- [CC25] Carmona Jiménez, J.L., Castrillón López, M. *Canonical reductive decomposition of extrinsic homogeneous submanifolds*. Note Mat. 45:(Suppl. 1) (2025), pp. 117–128. doi: [10.1285/i15900932v45s1p117](https://doi.org/10.1285/i15900932v45s1p117)
- [CCD25] Carmona Jiménez, J.L., Castrillón López, M., Díaz Ramos, J.C. *The Ambrose–Singer Theorem for cohomogeneity one Riemannian manifolds*. Transformation Groups (2025). DOI: [10.1007/s00031-025-09927-x](https://doi.org/10.1007/s00031-025-09927-x)
- [CC22] Carmona Jiménez, J.L., Castrillón López, M. *The Ambrose–Singer Theorem for General Homogeneous Manifolds with Applications to Symplectic Geometry*. Mediterr. J. Math. 19, 280 (2022). DOI: [10.1007/s00009-022-02197-x](https://doi.org/10.1007/s00009-022-02197-x)
- [CC22*] Carmona Jiménez, J.L., Castrillón López, M. *The homogeneous holonomies of complex hyperbolic space*. Ann. Glob. Anal. Geom. 62, 391–411 (2022). DOI: [10.1007/s10455-022-09852-2](https://doi.org/10.1007/s10455-022-09852-2)

[CC20] Carmona Jiménez, J.L., Castrillón López, M. *Reduction of Homogeneous Pseudo-Kähler Structures by One-Dimensional Fibers*. Axioms. 9(3):94 (2020). DOI: [10.3390/axioms9030094](https://doi.org/10.3390/axioms9030094)

PREPRINT:

[Car26*] **In preparation¹**. Carmona Jiménez, J.L. *On Generalize Imaginary Spin^c-Killing spinors*.
 [CCL26] **In preparation²**. Carmona Jiménez, J.L., Castrillón López, M., Loiudice, E. *The Ambrose-Singer Theorem for non-regular cohomogeneity one Riemannian manifolds*.

TEACHING EXPERIENCE

AT UNIVERSITY:

2023-2024	Riemannian Geometry I taught 6 hours of exercise sessions in Riemannian Geometry at the University of Bucharest for students of the Master of Science in mathematics.	University of Bucharest
2020 - 2022	Linear algebra, 1st year I taught 90 hours of exercise sessions in Linear algebra at Faculty of Mathematics and Faculty of Informatics in University Complutense of Madrid for two years.	University Complutense of Madrid
2020	Seminar Course. A seminar course of 10 hours entitled "The Ambrose-Singer Theorem". The Seminar course occurs during my stay at Faculty of Mathematics in Santiago de Compostela.	University of Santiago de Compostela

AT HIGH SCHOOL:

2022 - 2024 **Full teacher at secondary and upper-secondary School for two years.**
 I was teaching to students in I.E.S. Las Lagunas in Mijas, Spain and I.E.S. Mar de Alborán in Estepona, Spain.

LONG RESEARCH STAY

From: 15/09/2025 To: 30/11/2025	University of Marburg // Philipps-Universität Marburg Postdoctoral Research stay with Dr. Eugenia Loiudice at the Faculty of Mathematics and Informatics.	Germany
From: 01/05/2022 To: 31/07/2022	University of Marburg // Philipps-Universität Marburg Predoctoral research stay with Prof. Dr. habil. Ilka Agricola at the Faculty of Mathematics and Informatics.	Germany
From: 20/09/2021 To: 20/12/2021	University of Santiago de Compostela. Predoctoral research stay with Profesor Catedrático José Carlos Díaz Ramos at the department of Geometry and Topology.	Spain

SHORT RESEARCH STAYS

From: 23/06/2022 To: 27/06/2022	University of Marburg // Philipps-Universität Marburg. Research stay with Dr. Eugenia Loiudice at the Faculty of Mathematics and Informatics.	Germany
From: 17/10/2024 To: 28/10/2024	University of Málaga. Short research stay with Profesor Marián Cañadas at the department of Geometry and Topology. I gave a talk entitled: "On the history of Ambrose-Singer Theorems."	Spain

¹ This manuscript is in the final stages of preparation and drafts are available upon request.

² This manuscript is in the final stages of preparation and drafts are available upon request.

PROJECTS

From: 01/09/2025 To: 31/08/2027	Geometric structures on manifolds, symmetry and field theory My contribution: Participant. Principal researcher: Marco Castrillón López. Funding entity: Ministry of Science and Innovation. Start-end date: 01/09/2025 - 31/08/2027. Funding amount: 91.250,00 €	PID2024-156578NB-I00
From: 01/01/2024 To: 30/06/2026	Conformal Aspects of Geometry and Dynamics. My contribution: Participant. Principal researcher: Andrei Moroianu. Funding entity: <i>NextGenerationEU</i> , through the <i>Planul Național de Redresare și Reziliență</i> , Romania. Start-end date: 01/01/2024 - 31/06/2026. Funding amount: 6,000,000 RON ~ 1,200,000 €.	PNRR-III-C9-2023-I8 grant CF 149/31.07.2023
From: 01/09/2024 To: 31/08/2026	Geometrical, topological and combinatorial structures associated with dynamical systems. My contribution: Participant. Principal researcher: Marco Castrillón López and Luis Giraldo. Funding entity: Ministry of Science and Innovation. Start-end date: 01/09/2022 - 31/08/2025. Funding amount: 108,900.00 €.	PID2021-126124NB-I00
From: 01/01/2019 To: 31/12/2021	Combinatorics and dynamics: geometric and topological aspects in manifolds. Contribution: Participant. PIs: Marco Castrillón López and Luis Giraldo. Funding entity: Ministry of Science and Innovation. Start-end date: 01/01/2019 - 31/12/2021. Funding amount: 68,002.00 €.	PGC2018-098321-B-I00
2020	Algebra and geometry: Differential Geometry, Symplectic Geometry and Geometric Mechanics. I was a participant during my research contract at ICMAT.	20205-CEX001

GRANTS

From: 01/10/2020 To: 31/08/2022	FPI predoctoral fellowship at University Complutense of Madrid. Funding entity: Ministry of Science and Innovation, Spain. Start-end date: 01/10/2020 - 31/08/2022. Department: Álgebra, Geometry and Topology. Group of research: Geometrical, topological and combinatorial structures associated with dynamical systems. Principal researcher: Marco Castrillón López.	FPI
From: 6/11/2017 To: 6/07/2018	Research fellowship to collaborate with a department during the Master of Science in advanced mathematics at University Complutense of Madrid. Funding entity: Ministry of Science and Innovation, Spain. Start-end date: 06/11/2017 - 06/07/2018. Department: Department of Álgebra, Geometry and Topology at University Complutense of Madrid. Advisor: Marco Castrillón López.	

CONFERENCE PARTICIPATION

01/07/2025 Attendee.	Conformal Geometry and More – A conference in honor of Liviu Ornea. Facultatea de Matematică și Informatică, Bucharest, Romania. July 2025. Participation: Attendee.	
01/02/2025 Attendee.	MAM III: Quaternion-Kähler manifolds and the Lebrun-Salamon Conjecture. Philipps-Universität Marburg, Marburg, Germany. Participation: Attendee.	
01/01/2025 Speaker.	RSME's 7th Congress of Young Researchers. Real Sociedad Matemática Española, Bilbao, Spain. Participation: Speaker with talk entitled <i>Symplectic invariant connections</i> .	

01/10/2024 Speaker.	The 2nd International Conference on Differential Geometry-ICDG-Fez 2024. Fez, Morocco. Participation: Speaker with talk entitled <i>Cohomogeneity one invariant connections</i> .
01/08/2022 Speaker.	Conference Differential Geometry and its Applications. University of Hradec Králové. Participation: Speaker with talk entitled <i>The Ambrose-Singer Theorem for symplectic geometry</i> .
15/10/2021 Speaker.	Symmetry and Shape II. University of Santiago de Compostela. Participation: Speaker with talk entitled <i>On reductive locally homogeneous Fedosov manifolds</i> .
01/08/2021 Speaker.	Online Summer School on Geometry and Topology. University of Hradec Králové. Participation: Speaker with talk entitled <i>Fedosov Homogeneous Structures</i> .
01/12/2020 Speaker.	Third BYMAT conference: Bringing Young Mathematicians Together. Instituto de ciencias matemáticas (ICMAT). Participation: Speaker with talk entitled <i>The Ambrose- Singer Theorem for general homogeneous manifolds with applications to symplectic geometry</i> .
31/10/2019 Poster.	Symmetry and Shape I. University of Santiago de Compostela. Participation: Poster entitled <i>Characterization of simply-connected homogeneous spaces with geometric structures</i> .
01/09/2019 Poster.	Conference Differential Geometry and its Applications. University of Hradec Králové. Participation: Poster entitled <i>A generalization of Ambrose-Singer Theorem</i> .
08/07/2019 Speaker.	13th ICMAT International Summer School on Geometry, Mechanics and Control. Instituto de ciencias matemáticas (ICMAT), Madrid. Participation: Speaker with talk entitled <i>Locally homogeneous Riemannian spaces</i> .
24/05/2019 Attendee.	Second BYMAT conference: Bringing Young Mathematicians Together. Instituto de ciencias matemáticas (ICMAT), Madrid. Participation: Attendee.
01/07/2018 Attendee.	12th ICMAT International Summer Shool on Geometry, Mechanics and Control. University of Santiago de Compostela, Santiago. Participation: Attendee.

SEMINAR AND SHORT TALKS

24/06/2025	Geometry Seminar at Philipps-Universität Marburg Talk in English at Philipps-Universität Marburg Title: <i>The Ambrose-Singer Theorem for Cohomogeneity one Riemannian manifolds</i> .
11/04/2025	Geometry Seminar IMAR Talk in English at IMAR Title: <i>The Ambrose-Singer Theorem for cohomogeneity one manifolds</i>
21/10/2024	Lecture at the University of Málaga Talk in Spanish at UMA Title: <i>On the history of Ambrose-Singer Theorems</i>
15/10/2024	Geometry Seminar IMAR Talk in English at IMAR Title: <i>Ambrose-Singer theorems III</i>
08/10/2024	Geometry Seminar IMAR Talk in English at IMAR Title: <i>Ambrose-Singer theorems II</i>
01/10/2024	Geometry Seminar IMAR Talk in English at IMAR Title: <i>Ambrose-Singer theorems</i>
03/07/2023	Divulgation talk at VII Ph.D.ay Mathematics. Talk in Spanish at University Complutense of Madrid. Title: <i>Teselaciones y espacios homogéneos</i> . Title in English: <i>Tessellations and homogeneous spaces</i> .
12/04/2023	Round table about professional outings in Mathematics. Participant of the round table talking about my personal experience. Title: <i>Mesa Redonda de la IV Jornada sobre Salidas Profesionales de las Matemáticas</i> . Title in English: <i>Round Table of the IV Conference on Career Opportunities in Mathematics</i> .

04/04/2022	Interesting seminar of Geometry and Topology (UMA). Talk in English for the department of Geometry and Topology at University of Málaga. Title: <i>The Ambrose-Singer Theorem</i> .
24/06/2021	Divulcation talk at V Ph.D.ay Mathematics. Talk in Spanish at University Complutense of Madrid. Title: <i>Las geometrías homogéneas del espacio hiperbólico complejo</i> . Title in English: <i>The Homogeneous Geometries of Complex Hyperbolic Space</i> .
17/11/2021	Divulcation Seminar at University of Santiago de Compostela. Talk in Spanish. Title: Teselaciones y Espacios homogéneos. Title in English: Tessellations and homogeneous spaces.
02/07/2019	Divulcation talk at III Ph.D.ay Mathematics. Talk in Spanish at University Complutense of Madrid. Title: <i>El teorema de Ambrose-Singer. Un puente entre el Álgebra, la Geometría y el Análisis</i> . Title in English: <i>The Ambrose-Singer Theorem: A Bridge between Algebra, Geometry, and Analysis</i> .

LANGUAGES

English - Advanced. Spanish – native

TECHNICAL SKILLS

- **Computational Mathematics:** Python (NumPy, SymPy), SageMath (Symbolic Algebra, Tensor Calculus, Lie Algebras).
- **Tools:** LaTeX (Advanced typesetting), Git/GitHub, HTML.